



Review article

Long-term exposure to PM_{2.5} and stroke: A systematic review and meta-analysis of cohort studies

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ABSTRACT

Background: Stroke is one of the world's leading causes of death. Many studies have checked the relationship between short-term exposure to particulate matter (PM) and stroke, but few have focused on the effect of long-term exposure to PM_{2.5} (particulate matters with an aerodynamic diameter of $\leq 2.5 \mu\text{m}$). This study aimed to quantitatively examine the relationship of long-term exposure to PM_{2.5} with stroke incidence and mortality.

Methods: We identified relevant studies by searching the PubMed, EMBASE and MEDLINE. After the systematic review of pertinent studies, random-effect meta-analysis was conducted to investigate the association between long-term exposure to PM_{2.5} and stroke.

Results: Our meta-analysis included 16 cohort studies with more than 2.2 million people and above 49 149 endpoint events (incident stroke and death from stroke). The pooled hazard ratio (HR) for each $5 \mu\text{g}/\text{m}^3$ increment in PM_{2.5} was 1.11 (95% CI: 1.05, 1.17) (CI for confidence interval) for incidence of stroke and 1.11 (95% CI: 1.05, 1.17) for mortality of stroke. In the region-specific analysis, significant association between PM_{2.5} and incidence of stroke was found in North America (HR = 1.09, 95% CI: 1.05, 1.14) and Europe (HR = 1.07, 95% CI: 1.05, 1.10), while the pooled result of Asia showed no significance (HR = 2.31, 95% CI: 0.49, 10.95).

Conclusions: Long-term exposure to PM_{2.5} is an important risk factor for stroke. Since air quality is intimately related to everyone, policies aimed at reducing particulate matters will benefit public health a lot.

Declarations of interest

None.

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1. Introduction

Along with neonatal disorders and ischemic heart disease, stroke is the leading cause of disability-adjusted life-years (DALYs) globally and

is the third leading cause of years of life lost (YLLs) (2017) (GBD 2017 Causes of Death Collaborators, 2018; GBD 2017 DALYs and HALE Collaborators, 2018). The risk factors of stroke can be simply divided into two parts, modifiable (e.g., hypertension, diabetes) and non-modifiable risk factors (e.g., race, age). Identification and understanding the underlying mechanisms of modifiable risk factors can contribute to the prevention of stroke. Among those modifiable risk factors, air pollution has captured a lot of researchers' attention. Many kinds of air pollutants have been proved to be related to cardiovascular diseases. A meta-analysis of 94 articles (Shah et al., 2015) revealed a positive association between short-term exposure to carbon monoxide and nitrogen dioxide and stroke. The same is true for PM_{2.5} and PM₁₀ (particulate matters with an aerodynamic diameter of $\leq 2.5 \mu\text{m}$ and

Abbreviations: PM_{2.5}, particulate matters with an aerodynamic diameter of $\leq 2.5 \mu\text{m}$; PM₁₀, particulate matters with an aerodynamic diameter of $\leq 10 \mu\text{m}$; HR, hazard ratio; OR, odds ratio; CI, confidence interval; DALYs, disability-adjusted life-years; YLLs, years of life lost; IQR, interquartile range; ICD, International Classification of Diseases

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$\leq 10 \mu\text{m}$). Compared with short-term exposure, relationship between long-term exposure to air pollution and stroke is controversial. A prospective cohort study performed in six low- and middle-income countries (Lin et al., 2017) indicated that ambient $\text{PM}_{2.5}$ was a significant risk factor of stroke incidence (OR = 1.12, 95% CI: 1.04, 1.21) (OR for odds ratio, CI for confidence interval). An America-nationwide cohort studies of nurses (Hart et al., 2015) during 1989–2006 reported no statistically significant association between $\text{PM}_{2.5}$ and incident stroke (HR = 1.03, 95% CI: 0.92, 1.15) (HR for hazard ratio). A meta-analysis in 2015 (Scheers et al., 2015) indicated a positive association between long-term PM_{10} exposure and stroke in North America.

Previous evidence reported an increased harm effect with a decreased size of particulate matter (Delfino et al., 2005; Ge et al., 2018). But currently the monitoring data of ultrafine particles is not available in many regions. Therefore, our studies focused on the impact of $\text{PM}_{2.5}$ among healthy populations. Besides, details about the stroke type, different susceptibility of population, interaction with smoking status will also be discussed in this work. A better understanding of the relationship between air pollution and stroke may offer guidance to public health management and environmental policy.

2. Methods

This work was performed according to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines (<http://www.prisma-statement.org/>), and it was registered in the International Prospective Register of Systematic Reviews (PROSPERO) (registration number: CRD42019124093).

2.1. Literature search

We identified relevant studies published from 1980 to December 2018 by searching on PubMed, EMBASE and MEDLINE. Studies were restricted to human species and English language. We used the keywords “Stroke” in combination with “Particulate Matter” or “Air Pollution”. The full electronic search strategy for PubMed is: (((((((((((Stroke[MeSH Terms]) OR Strokes[Title/Abstract]) OR Cerebrovascular Accident[Title/Abstract]) OR Cerebrovascular Accidents[Title/Abstract]) OR CVA[Title/Abstract]) OR CVAs[Title/Abstract]) OR Apoplexy[Title/Abstract]) OR Vascular Accident, Brain [Title/Abstract]) OR Brain Vascular Accident[Title/Abstract]) OR Brain Vascular Accidents[Title/Abstract]) OR Vascular Accidents, Brain [Title/Abstract]) OR stroke[Title/Abstract])) AND (((((((((((Air Pollution[MeSH Terms]) OR Air Pollution[Title/Abstract]) OR Air Pollution[Title/Abstract]) OR Pollution, Air[Title/Abstract]) OR Air Quality[Title/Abstract]) OR Particulate Matter[MeSH Terms]) OR Particulate Matter[Title/Abstract]) OR Airborne Particulate Matter [Title/Abstract]) OR Particulate Matter, Airborne[Title/Abstract]) OR Air Pollutants, Particulate[Title/Abstract]) OR Particulate Air Pollutants[Title/Abstract]) OR Pollutants, Particulate Air[Title/Abstract]) OR Ambient Particulate Matter[Title/Abstract]) OR Particulate Matter, Ambient[Title/Abstract])).

No filter for study designs was applied. Pertinent studies were also identified during the screen of reviews and reference lists of search results. After the exclusion of duplications, titles and abstracts were screened during the initial selection. Further review was based on the full text. Yuan S and Wang JX screened all the articles independently. Their results reached consensus. A third reviewer (He ZY) would check the article and make decisions if there was disagreement.

2.2. Inclusion and exclusion criterion

Cohort studies were included if they examined the relationship between long-term exposure to $\text{PM}_{2.5}$ and incidence or mortality of stroke among healthy populations. The exclusion criterion included: 1. Duplications or same cohorts with similar study period, unless they had

different endpoints. We keep the studies with explicit $\text{PM}_{2.5}$ information when dealt with duplications. 2. Reviews, meta-analysis or case reports. 3. Studies which did not provide clear hazard ratio or odds ratio with 95% confidence interval, and could not retrieve by contact with the authors. 4. Studies which restricted the population to specific patients, such as patients with cardiovascular disease at the start.

2.3. Data extraction

Data extraction was carried out by two reviewers (Yuan S and Wang JX) independently. From each eligible study, we collected author name, publication year, geographical region, study period, total sample size, gender ratio, endpoints, definition of stroke, mean concentration of $\text{PM}_{2.5}$, time window of exposure, HRs or ORs with 95% CI and subgroups information. When there were results of different calculation models, we extracted the HR fully adjusted for covariates.

2.4. Quality assessment

Used the previous similar study (Scheers et al., 2015) and Newcastle-Ottawa Quality Assessment Scale (NOS) (http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp) as reference, we assessed the quality of the eligible studies from 5 aspects: study design, characteristic of the cohorts, adjusted model for HR calculation, definition of stroke and the estimation of $\text{PM}_{2.5}$ (Table I, please see supplemental material).

2.5. Data standardization

Since the interquartile range (IQR) of $\text{PM}_{2.5}$ was different in each study, we recalculated the HR and CI by adjusting the IQR to $5 \mu\text{g}/\text{m}^3$ with the following formula:

$\text{HR}_{\text{adjusted}} = \text{HR}_{\text{primary}}^{\circ(5/\text{IQR})}$. ($\text{HR}_{\text{adjusted}}$: The adjusted HR value; $\text{HR}_{\text{primary}}$: The primary HR value from included studies).
The conversion of CI is analogous.

2.6. Statistical analysis

Random-effects model was used to calculate the pooled HR and 95% CI. Heterogeneity was tested with the I^2 statistic. We combined Funnel plot asymmetry and Egger's regression to detect publication bias, and defined significant publication bias as a p value < 0.1. Sensitivity analysis was used to examine the stability of the pooled results, whereby each study was omitted at a time and recalculated the pooled HR of the rest studies. We also performed subgroup analysis by geographic region, gender, stroke type and smoking status to examine details about the association. Meta-analysis was performed with Stata software (version 12.0; College station, TX, USA). All p values were two-side with a significant level at 0.05.

3. Results

A total of 999 studies were included in our study, including the results of keywords search ($n=997$) and additional records identified from reference lists of other studies ($n=2$). After the detailed reading of full texts, 27 articles were potentially eligible for meta-analysis. Finally, a total of 16 cohort studies were included in our analysis (Fig. 1).

Of those studies, 10 studies reported the relationship between long-term exposure to $\text{PM}_{2.5}$ and incidence of stroke (Cai et al., 2018; Gandini et al., 2018; Hart et al., 2015; Kim et al., 2017; Lin et al., 2017; Puett et al., 2011; Qiu et al., 2017; Stafoggia et al., 2014; Stockfelt et al., 2017; To et al., 2015), 4 studies reported on mortality of stroke (Beelen et al., 2009, 2014; Turner et al., 2016; Wong et al., 2015). 2 studies reported two endpoints (Lipsett et al., 2011; K. A. Miller et al., 2007). Cohorts of two studies (Beelen et al., 2014; Stafoggia et al., 2014) were overlapped but reported on incidence and mortality of

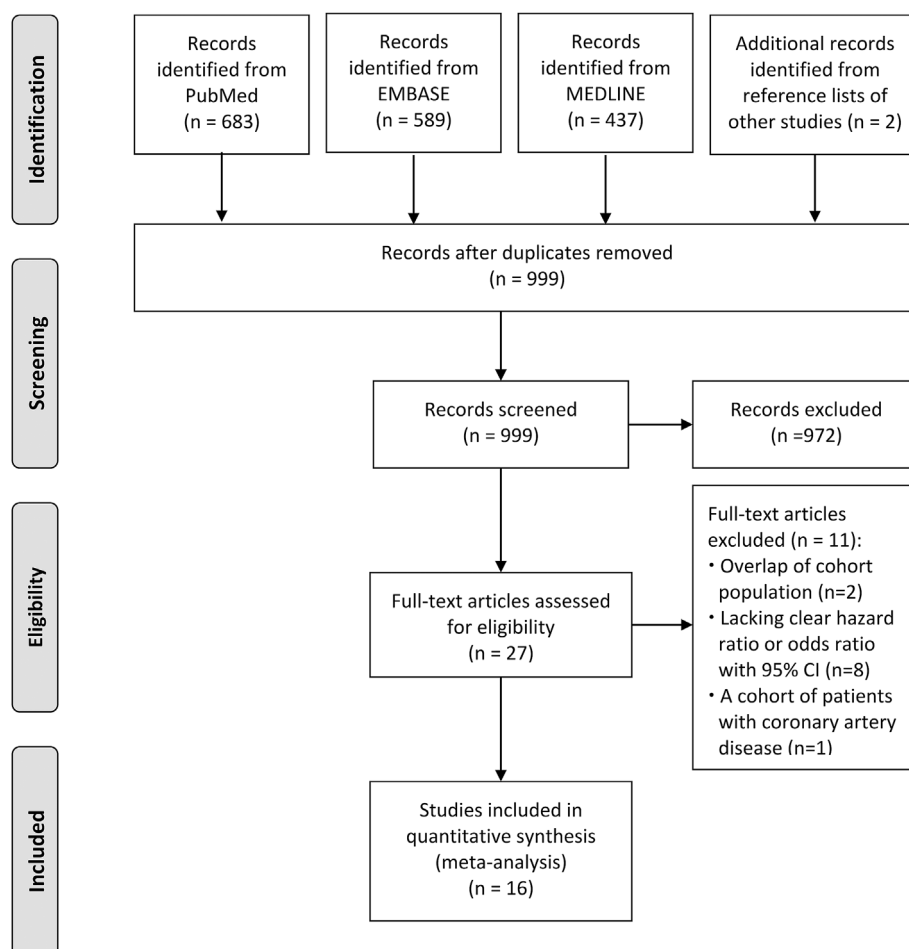


Fig. 1. Results of systematic literature search.

stroke respectively, therefore they were both included. The main information of eligible studies was given in Table 1. We assessed the quality of included studies from 5 aspects and their scores were no less than 7 points (total score=12.5 points) (Table I, please see supplemental material). Part of primary information about the quality assessment was listed in Table II (please see supplemental material).

3.1. Risk of bias assessment for included studies

3.1.1. Bias from character of population

Cohorts of 12 included studies are representative of the average population, whereas other 4 studies restricted their cohorts to specific population, such as female public-school professionals (Lipsett et al., 2011), postmenopausal women (K. A. Miller et al., 2007), male health professionals (Puett et al., 2011) and nurses (Hart et al., 2015). More than half of included studies have not given clear demonstration that stroke was not present at the start of study, while 2 studies presented explicit results with and without endpoints happened within one year (Qiu et al., 2017; Wong et al., 2015). This diversity of included population means potential different physical fitness, distinct occupational exposure and socioeconomic status, which may contribute to the bias of synthesized results.

3.1.2. Bias from exposure measurement

4 studies (Lin et al., 2017; Qiu et al., 2017; To et al., 2015; Wong et al., 2015) performed satellite-based estimates for exposure measurement, whereas 10 studies (Beelen et al., 2009, 2014; Cai et al., 2018; Gandini et al., 2018; Hart et al., 2015; Lipsett et al., 2011; Puett et al., 2011; Stafoggia et al., 2014; Stockfelt et al., 2017; Turner et al.,

2016) combined data from air pollution monitoring stations and interpolation model to calculate the exposure concentration of $PM_{2.5}$. Interpolation model was not mentioned in 2 studies (Kim et al., 2017; K. A. Miller et al., 2007). Besides, included studies have different time windows of exposure but no less than 1 year. 2 studies (Lin et al., 2017; Puett et al., 2011) found similar results among different time windows, such as 1, 2 and 4 years. 1 study (Hart et al., 2015) found higher HRs for the 1- and 2-year exposure times compared to the 5- and 10-year averages.

3.1.3. Bias from covariates adjustment

All 16 studies adjusted their results by age and sex. Only one study (Kim et al., 2017) did not adjust the result by smoking status. Since obesity is closely associated with cardiovascular morbidity and mortality (Khan et al., 2018), most studies have made adjustment for BMI, except 4 studies (Beelen et al., 2009; Cai et al., 2018; Stafoggia et al., 2014; Wong et al., 2015). Occupational exposure was taken into account in 7 studies (Beelen et al., 2014; Gandini et al., 2018; Puett et al., 2011; Stafoggia et al., 2014; Stockfelt et al., 2017; To et al., 2015; Turner et al., 2016). Hart et al. further adjusted the result for race, menopausal status and hormone use, hypercholesterolemia, hypertension, diabetes and family history of myocardial infarction (Hart et al., 2015).

3.1.4. Bias from definition of endpoints

Most included studies defined endpoint events according to International Classification of Diseases. 4 studies (Hart et al., 2015; Lin et al., 2017; K. A. Miller et al., 2007; Puett et al., 2011) did not give clear definition of stroke.

Table 1
Characteristics of the included studies.

Study	Area	Study period	Total sample size	Age (Years)	Gender, female (%)	PM _{2.5} Concentration (SD)	Time windows of exposure	Endpoints	No. of Cases	Definition of stroke
Beelen et al. (2009)	Netherlands	1987–1996	117528	55–69	51.7	NA	1987–1996	Mortality	1175	ICD-9 430–438
Beelen et al. (2014)	13 countries, Europe	1985–2012	367383	All	63.0	6.6–31.0 ^a	2008–2011	Mortality	2484	ICD-10 I60–I69
Lipsett et al. (2011)	California, USA	1999–2005	124614	≥ 30	100.0	15.64 ± 4.48	1999–2005	Incidence	969	ICD-9 430–438
Miller et al. (2007)	36 cities, USA	1994–2002	65893	50–79	100.0	13.5 ± 3.7	In 2000	Mortality	382	ICD-10 I60–I69
Puett et al. (2011)	13 states, USA	1989–2003	17545	40–75	0.0	17.8 ± 3.4	12 months	Incidence	554	ICD-10
Stafoggia et al. (2014)	5 countries, Europe	1992–2010	99446	All	54.2	7–31 ^a	2008–2011	Mortality	122	NA
To et al. (2015)	Canada	1980–2013	29549	40–59	100.0	12.62 ± 2.361	1998–2006	Incidence	300	NA
Wong et al. (2015)	Hong Kong, China	1998–2011	60221	≥ 65	65.0	26.4–44.6 ^a	2000–2011	Mortality	1621	ICD-9 431–436
Lin et al. (2017)	China, Ghana, India, Mexico, Russia and South Africa	2007–2010	45625	≥ 18	56.8	23.09 [†]	3 years	Incidence	1239	ICD-10 I60–I69
Kim et al. (2017)	Seoul, Korea	2007–2013	136094	≥ 18	50.9	25.03 ± 14.29	2007–2013	Incidence	934	ICD-10 I60–64
Qiu et al. (2017)	Hong Kong, China	1998–2010	61447	≥ 65	65.9	35.8 ± 2.4	1998–2010	Incidence	6244	ICD-9 430–436
Hart et al. (2015)	11 states, USA	1989–2006	96058	30–55	100.0	12.0 ± 2.8	2000–2006	Incidence	1194	NA
Cai et al. (2018)	the United Kingdom	1993–2010	289128	≥ 20 & 40–69	58.9	10.0 ± 1.1	2001–2006	Incidence	2871	ICD-9 430–438
Stockfelt et al. (2017)	Gothenburg, Sweden	1990–2011	10350	64–75 & 25–64	0 (PPS), 52.0 (GOT-MONICA)	9.3[6.3–12] [‡] 8.5[5.6–12]	2007–2011	Incidence	1391	ICD-10 I60–I69
Turner et al. (2016)	The United States	1982–2004	669046	≥ 30	56.2	12.6 ± 2.9	1999–2004	Mortality	17 085	ICD-9 430–438
Gandini et al. (2018)	Italy	1999–2008	74989	> 35	52.7	20.5–33.4 ^{a†}	1999, 2003, 2005, 2007	Incidence	1505	ICD-10 I60–I69

Abbreviations: PM_{2.5}, Particulate matters with an aerodynamic diameter of ≤ 2.5 μm; SD, Standard deviation; ICD, International Classification of Diseases; NA, Not applicable; PPS and GOT-MONICA are the 2 cohorts which are described respectively by Stockfelt et al. (Stockfelt et al., 2017) in their study.

^a Only PM_{2.5} Concentration range was available; [†]SD was absent but the 5th–95th percentiles were subsistent.

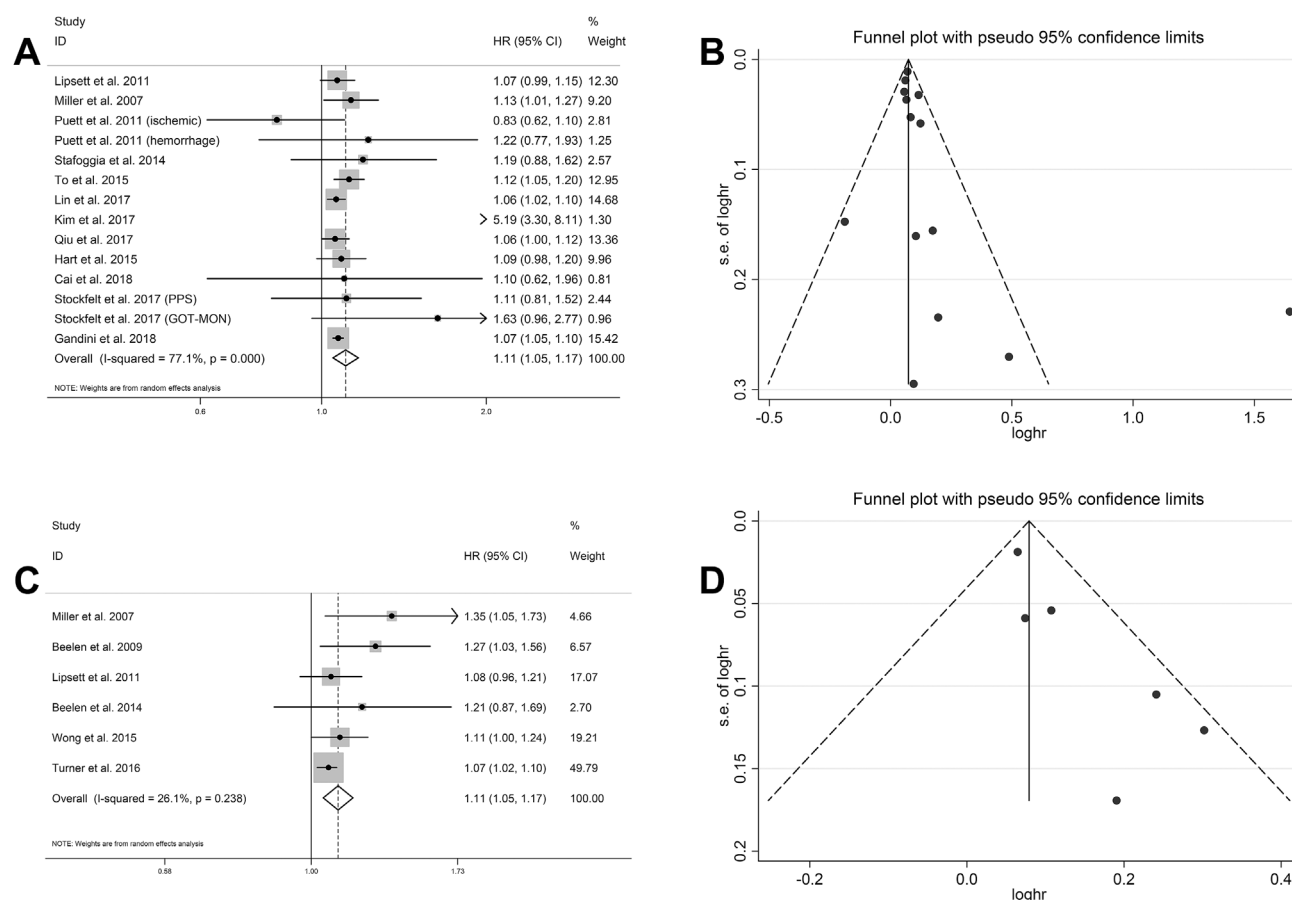


Fig. 2. A $PM_{2.5}$ and incident stroke. There is no overall result of $PM_{2.5}$ and stroke incidence in the study of Puett et al. (2011). Therefore, we select the hazard ratios of both ischemic and hemorrhage stroke for the meta-analysis. PPS and GOT-MONICA are the 2 cohorts which are described respectively by Stockfelt et al. (2017) in their study. B Funnel blot is applied to analyze the publication bias of $PM_{2.5}$ and stroke incidence. C Long-term exposure to $PM_{2.5}$ and mortality of stroke. D Funnel blot is employed to estimate the publication bias of $PM_{2.5}$ and stroke mortality.

3.2. $PM_{2.5}$ and stroke incidence

Fig. 2A showed the HR of 12 studies about the $PM_{2.5}$ and incidence of stroke. The pooled hazard ratio for each $5 \mu g/m^3$ increment in $PM_{2.5}$ was 1.11 (95% CI: 1.05, 1.17), which indicated a positive association between long-term exposure to $PM_{2.5}$ and incidence of stroke. The heterogeneity of these studies was significant ($I^2 = 77.1\%$, $p = 0.000$). Funnel plot and Egger's regression found no evidence for publication bias (Egger's regression: $p = 0.140$, Fig. 2B). Sensitivity analysis indicated the results were statistically robust (Figure IA, please see supplemental material).

The subgroup-analysis by gender suggested significant association both in males and females (Table 2) (Figure IIA, please see

supplemental material). Pooled results of 4 studies found no association between $PM_{2.5}$ and incidence of hemorrhage or ischemic stroke (Table 2) (Figure IIB, please see supplemental material). The region-specific analysis suggested a positive association in North America (HR: 1.09, 95% CI: 1.05, 1.14) and Europe (HR: 1.07, 95% CI: 1.05, 1.10) (Table 2) (Figure IIC, please see supplemental material). But the pooled result in Asia (only 2 studies) showed a high heterogeneity ($I^2 = 97.9\%$, $p = 0.000$). When we excluded the two Asian studies and pooled results of North America and Europe together, the heterogeneity of pooled HR was no longer significant (HR = 1.08, 95% CI: 1.06, 1.10; $I^2 = 0.0\%$, $p = 0.551$) (Figure IID, please see supplemental material).

For individuals who never smoke and who have ever smoked, similar significant association between long-term exposure to $PM_{2.5}$ and

Table 2
Subgroup-analysis of $PM_{2.5}$ and stroke incidence.

Subgroup	Stratum	Number of studies	Pooled HR (95%CI)	Test of heterogeneity	
				I^2 (%)	p value
Gender	Male	6	1.08 (1.05, 1.10)	0.0	0.795
	Female	9	1.07 (1.04, 1.10)	4.0	0.401
Stroke type	Ischemic stroke	4	1.62 (0.88, 2.97)	93.9	0.000
	Hemorrhagic stroke	4	1.20 (0.79, 1.80)	64.2	0.039
Region	North America	5	1.09 (1.05, 1.14)	3.6	0.394
	Europe	4	1.07 (1.05, 1.10)	0.0	0.578
	Asia	2	2.31 (0.49, 10.95)	97.9	0.000

Abbreviations: $PM_{2.5}$, Particulate matters with an aerodynamic diameter of $\leq 2.5 \mu m$; HR, Hazard ratio. CI, Confidence interval. I^2 is the value of total variation clarified by heterogeneity.

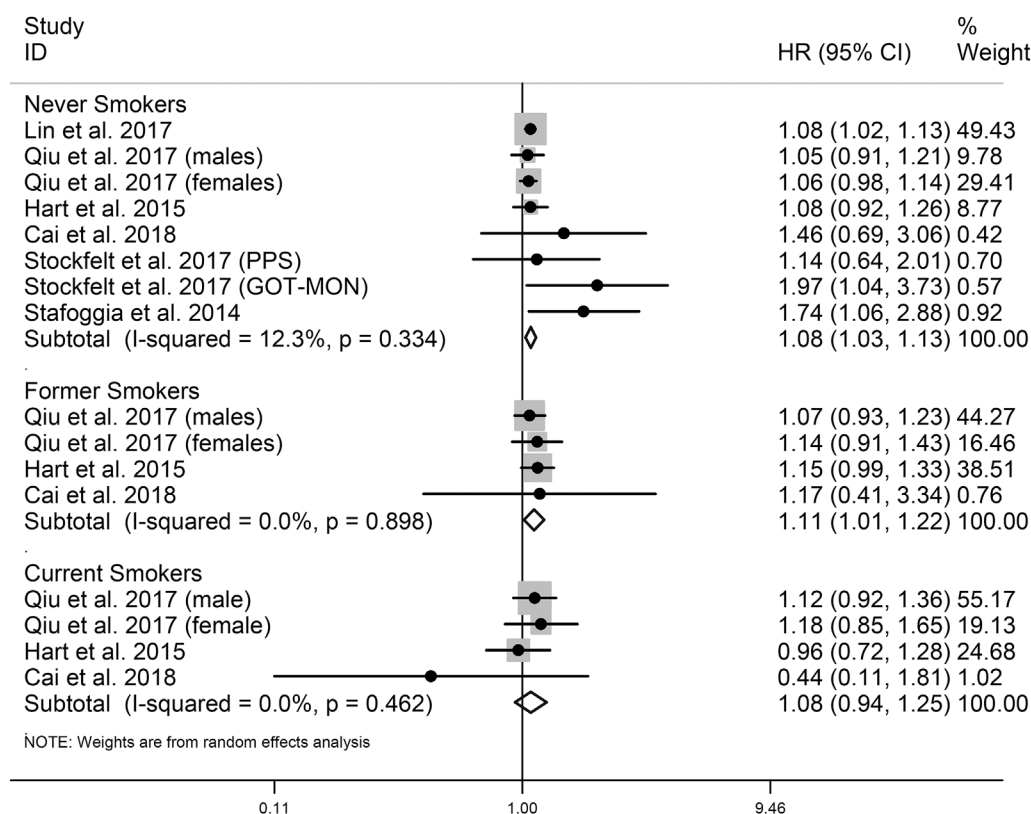


Fig. 3. Subgroup-analysis (by smoking status) of PM_{2.5} and incident stroke. Qiu et al. (2017) performed the subgroup-analysis among males and females respectively, which are both included in our meta-analysis. PPS and GOT-MONICA are the 2 cohorts which are described respectively by Stockfelt et al. in their study.

incidence of stroke was found (never smokers: HR = 1.08, 95% CI: 1.03, 1.13; ever smokers: HR = 1.06, 95% CI: 1.01, 1.10). When we subdivided ever-smokers into former and current smokers, the pooled analysis indicated a positive association among former smokers (HR = 1.11, 95% CI: 1.01, 1.22), whereas the association was not significant among current smokers (HR = 1.08, 95% CI: 0.94, 1.25) (Fig. 3).

3.3. PM_{2.5} and mortality of stroke

We included 6 studies to examine the association between long-term exposure to PM_{2.5} and mortality of stroke (Fig. 2C). A random-effect meta-analysis suggested a similar positive result to that of stroke incidence, with a HR of 1.11 (95% CI: 1.05, 1.17). The heterogeneity of these studies was not significant ($I^2 = 26.1\%$, $p = 0.238$). Funnel plot and Egger's regression found significant publication bias (Egger's regression: $p = 0.023$, Fig. 2D). Sensitivity analysis indicated the results were statistically robust (Figure IB, please see supplemental material). The stratified analysis by gender revealed positive association among females (HR = 1.12, 95% CI: 1.01, 1.24), but not males (HR = 1.15, 95% CI: 0.96, 1.37) (Figure IIIA, please see supplemental material). The region-specific analysis found positive association both in the North America (HR = 1.09, 95% CI: 1.01, 1.18) and Europe (HR = 1.26, 95% CI: 1.05, 1.50) (Figure IIIB, please see supplemental material). Only one study reported the relationship between PM_{2.5} and mortality of stroke in Asia with a HR of 1.11 (95% CI: 1.00, 1.24).

4. Discussion

Our meta-analysis has quantitatively examined the association between long-term exposure to PM_{2.5} and stroke. The pooled analysis included 16 cohort studies with more than 2.1 million people and about 49149 endpoint events (incident stroke and death from stroke). The results show that long-term exposure to PM_{2.5} is significantly associated

with both incidence of stroke and mortality of stroke. The region-specific analysis found significant association between PM_{2.5} and incidence of stroke in North America and Europe. Only two included studies reported relevant cohort in Asia and the pooled result shows great heterogeneity. A meta-analysis conducted in 2015 found similar results between long-term exposure to PM₁₀ and incidence of stroke (Scheers et al., 2015).

The subgroup analysis by gender reveals significant association between PM_{2.5} and incidence of stroke in both males and females. When we used mortality of stroke as endpoint, the pooled results indicate positive association in females but not in males. Previous studies also reported a significant association between long-term exposure to particulate matter and mortality of cardiovascular diseases among females but not males (Chen et al., 2005; Franklin et al., 2007). Females after disorder in pregnancy have higher risk of stroke and hysterectomy is a possible protective factor (Poorthuis et al., 2017). Further understanding of the gender-specific risk of stroke can contribute to the precise health assessment.

The stratified analysis by stroke type finds no significant association between long-term exposure to PM_{2.5} and incidence of ischemic or hemorrhage stroke. Many previous studies reported positive association between particulate matter and ischemic stroke (Maheswaran et al., 2012; Tsai et al., 2003; Yorifuji et al., 2013), but the results of hemorrhage stroke were inconsistent. More works are needed to detect the different risk of ischemic and hemorrhage stroke and their mechanism.

4.1. Biological mechanisms

A number of studies have revealed that air pollutants, especially particulate matter, could promote the occurrence of cardiovascular events (Brook et al., 2004; Brook and Rajagopalan, 2010; Laden et al., 2006). Compared with short-term exposure, population with long-term

exposure to high levels of PM have a higher incidence and mortality of cardiovascular diseases (Yin et al., 2017). The specific biological mechanism of long-term exposure to particulate matters is still evolving but can be roughly summarized as two steps, primary initiating pathway and secondary effector pathway (Rajagopalan et al., 2018). As the primary initiating pathway, Oxidative stress (Araujo and Nel, 2009), direct translocation (M. R. Miller, 2017) and biologic intermediates in response to exposure (Poschl and Shiraiwa, 2015) mediate a variety of secondary effects in long-term exposure, including endothelial dysfunction, systemic inflammatory response, thrombotic pathways, autonomic imbalance, activation of hypothalamic and pituitary-adrenal axis (HPA) and epigenomic changes (Rajagopalan et al., 2018). The above pathways interact with each other and are the key factors in the incidence of cardiovascular diseases. Similarly, they also play a crucial role in the occurrence of cerebrovascular diseases, especially ischemic stroke. Current studies have suggested that PM_{2.5} could cause blood-brain barrier dysfunction and enter the central nervous system, inducing direct or indirect inflammatory reactions, which may be related to the occurrence of cerebrovascular diseases or stroke (Block and Calderon-Garciduenas, 2009). In addition, PM_{2.5} has been confirmed to regulate the expression of genes involved in key vascular regulatory pathways in the brain, which reflexes the opinion that inhaled pollutants can induce cerebrovascular effects (Thomson et al., 2007).

4.2. Strengths and limitations

The 16 studies we included for meta-analysis are all cohort studies. We did not include ecological studies, because most of them lack the baseline information of the population and are difficult to adjusted their hazard ratios by covariates such as body mass index (BMI) and smoking status.

With the stratified analysis by smoking status, we found a significant association between long-term exposure to PM_{2.5} and incident stroke among never smokers and former smokers. The results among current smokers was not significant. This result has not been reported before. First, smoking is an independent risk factors of cardiovascular disease (Doll et al., 2004; GBD 2017 Risk Factor Collaborators, 2018; Thun et al., 2013). The impact of PM_{2.5} may be covered by smoking. For example, in one of the included studies (Qiu et al., 2017), we found that although the relationship between stroke incidence and long-term exposure to PM_{2.5} among current smokers was not significant, the incidence of stroke was still higher among smokers than never smokers. A life course analysis of the Framingham Heart Study revealed that over the entire lifetime male non-smokers had higher risks of stroke, while the risk of cardiovascular disease before age 70 was higher among smokers (Mamun et al., 2004). Shorten of life caused by smoking accounted for this result. Whether the shorten of life expectancy can also account for the impact of smoking status on the relationship between PM_{2.5} and stroke is not clear. Because the age-specific subgroups of individual study have different grouping criteria, we cannot pool the results of different age groups. Relatively high mortality among smokers may affect our examination of stroke incidence. Furthermore, many studies (Agbenyikey et al., 2011; Cameron et al., 2010; Nafees et al., 2012) have revealed that smoking is closely related to a high level of PM_{2.5} concentration around smokers. So, the concentration of PM_{2.5} used in our study cannot precisely reflect the real exposure to PM_{2.5}, especially among smokers. Besides, we must admit that the sample size of current smokers is not enough. To better understand the interaction between smoking, PM_{2.5} and stroke, more works with exact indoor concentration of PM_{2.5} and comprehensive analysis of all-cause mortality and stroke incidence remain to be done.

We admitted limitations in the following aspects. First, the number of included studies is not enough, especially for the mortality of stroke and subgroup-analysis. Second, the pooled result of PM_{2.5} and incident stroke has high heterogeneity, which disappeared after the exclusion of

two studies conducted in Asia. The pooled results in Asia have only 2 studies and high heterogeneity, partly because PM_{2.5} data of many regions in Asia has not been monitored. A recent estimate of global burden of disease attributable to ambient air pollution pointed out that south and east Asia contributed 59% of the 4.2 million global deaths attributable to ambient PM_{2.5} in 2015 (Cohen et al., 2017). A retrospective study in China indicated a positive association between long-term exposure to PM₁₀ and all-cause mortality, cardiovascular disease mortality and cerebrovascular disease mortality. To better understand the particle matter's impact on health in Asia, more works about air pollution and epidemiology remain to be done. Third, pollutant air is a heterogeneous mixture of gases, liquids, and particulate matters. Other air pollutants such as nitric oxide and ozone may have interaction with PM_{2.5} and strengthen their effect. We should examine and quantify a compounding effect of all these air pollutants on increasing the risk of stroke.

4.3. Recommendations for future research

4.3.1. Study design

We suggest high-quality longitudinal studies with detailed baseline information and regular health examination as ideal study method. Population with different features, such as a specific occupation or different age groups, should be recorded and synthesized by stratified analysis. Race, geographical region and other potential factors should be taken into account to improve the representativeness of the study.

4.3.2. Exposure measurement

Future studies should recalculate the data from air pollution monitoring stations by interpolation model to assess the level of exposure. Large-scale studies which include areas with less air pollution stations can complete the information by satellite-based estimates. Further, concentration of air pollutants should be measured during the entire study period.

4.3.3. Adjustment by covariates

During the review of eligible studies, we found various covariates which may exert unstable impact on the summary. We recommend standardized covariates included at least age, sex, BMI, smoking status, alcohol consumption, occupational exposure, hypertension, cholesterol and socioeconomic status at an individual level. Further adjustment can be performed by personal habits (such as physical activities, diet) and health status (such as diabetes, family history of cardiovascular disease). This work requires detailed and elaborate baseline information.

4.3.4. Definition of endpoints

The follow-up time should be long enough to observe the occurrence of endpoint events. The definition of stroke should follow the International Classification of Diseases. We recommend diagnosis by at least two physicians independently or diagnosis with explicit evidence of medical imaging. Self-reported endpoints from questionnaires are not recommended.

5. Conclusions

Our meta-analysis identified long-term exposure to PM_{2.5} as a significant risk factors of stroke. But more high-quality longitudinal studies are needed to understand details of this association, especially in Asia. Since almost all population in a region are exposure to air pollution, comprehensive particulate matter control policy is necessary to promote health and reduce the burden of stroke.

Disclosures

None.

Conflicts of interest

None declared.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.envres.2019.108587>.

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